PATTERNS:

1. FULL PYRAMID

```
public class pyramid {
public static void main(String[] args) {
   int n = 5;
   for (int i = 1; i <= n; i++)
   {
      for (int j = i; j < n; j++) {
         System.out.print(" ");
      }
      for (int k = 1; k <= (2 * i - 1); k++) {
         System.out.print("*");
      }

     System.out.println();
   }
}
}
```

2.INVERTED FULL PYRAMID

```
public class inverted
{
   public static void main(String[]args){
      int n=5;
      for(int i=n;i>=n;i--){
         for(int j=n;j>i;j--){
            System.out.print("");
            for(int k = 1; k <= (2 * i - 1); k++){
            System.out.print("*");
         }
         System.out.println();
         }
      }
   }
}
```

3.DIAMOND PATTERN

```
public class diamond{
   public static void main(String[]args){
      int n=5;
      for(int i=1;i<=n;i++)
      {
         for(int j=i;j<n;j++){
```

```
            System.out.print(" ");
        }
        for(int k=1;k<=(2*i-1);k++){
            System.out.print("*");
        }
        System.out.println();
    }
    for(int i=n-1;i>=1;i--){
        for(int j=i;j<n;j++){
            System.out.print(" ");
            }
        for(int k=1;k<=(2*i-1);k++){
            System.out.print("*");
        }
        System.out.println();
    }
  }
}
```

4.HOLLOW PYRAMID

```
public class hollow{
  public static void main(String[]args){
    int n=5;
    for(int i=1;i<=n;i++){
        for(int j=1;j<=n;j++){
            System.out.print(" ");
        }
        for(int j=1;j<=2*i-1;j++){
            if(j==1 ||j==2*i-1||i==n){
                System.out.print("*");
                    }else{
                        System.out.print(" ");
                    }
        }
        System.out.println();
    }
  }
}
```

5.HOLLOW INVERTED PYRAMID

```
public class invhollow{
  public static void main(String[]args){
    int n=5;
    for(int i=1;i>=n;i--){
        for(int j=1;j<=n-i;j++){
            System.out.print(" ");
```

```
        }
        for(int j=1;j<=2*i-1;j++){
           if(i==n||j==1||j==2*i-1){
              System.out.print("*");
                    }else{
                       System.out.print(" ");
                    }
        }
        System.out.println();
     }
   }
}
```

6. HOURGLASS PATTERN

```
public class hourglass{
   public static void main(String[]args){
      int n=5;
    for (int i = n; i >= 1; i--) {
        for (int j = 0; j < n - i; j++) {
           System.out.print(" ");
        }
        for (int j = 1; j <= 2 * i - 1; j++) {
           System.out.print("*");
        }
        System.out.println();
      }
      for (int i = 2; i <= n; i++) {
        for (int j = 0; j < n - i; j++) {
           System.out.print(" ");
        }
        for (int j = 1; j <= 2 * i - 1; j++) {
           System.out.print("*");
        }
        System.out.println();
   }
}
}
```

7.BUTTERFLY PATTERN

```
public class butterfly{
   public static void main(String[]args){
      int n=5;
      for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
           System.out.print("*");
        }
```

```
        for (int j = 1; j <= 2 * (n - i); j++) {
            System.out.print(" ");
        }

        for (int j = 1; j <= i; j++) {
            System.out.print("*");
        }

        System.out.println();
    }
     for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            System.out.print("*");
        }

        for (int j = 1; j <= 2 * (n - i); j++) {
            System.out.print(" ");
        }

        for (int j = 1; j <= i; j++) {
            System.out.print("*");
        }

        System.out.println();
    }
  }
}
```